

Preview

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Subject: Paper TPWRS-00958-2015 by Mr. Pratyasa Bhui, Real Time Prediction and Control of Transient Stability Using Transient Energy Function

Body: After careful review we have determined that your paper might be of interest for publication in the IEEE Transactions on Power Systems, after revisions, if those revisions fully and properly address the concerns of the reviewers. The comments from the reviewers appearing below may include specific revisions that are mandatory and other changes suggested. You may have to make broader changes if necessary based on general comments provided by the reviewers.

I wish to emphasize that this is not a conditional acceptance.

It is, rather, a recognition by our reviewers and editors that your paper merits some interest. The reviewers of the original draft will be approached to assess the revised manuscript; at the discretion of the editor, additional reviewers may be included.

Please submit your revised paper within 60 days of receiving this letter. After 60 days the revision will expire and you will need an extension in order to submit the revised paper.

We encourage you to include an explanation of how you responded to the comments by the reviewers to help expedite our review.

Please include this explanation in the space provided for "response to reviewers" and "response to editor" when you upload the revision.

Please note that you should submit your revised paper following the latest author's guidelines (We have had direct author upload since January 7, 2002). See the Author's Kit on:

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Also, directions are contained at the end of this email.

WE ARE AUTOMATICALLY INFORMED WHEN YOU UPLOAD YOUR REVISION AND RESPONSES SO PLEASE DO NOT SEND EMAIL TO ME THAT YOU SUBMITTED YOUR PAPER, AND PLEASE DO NOT SEND ME YOUR PAPER AS EMAIL ATTACHMENT.

Thanks,

Prof. Antonio Conejo, Editor in Chief,
Transactions on Power Systems

COMMENTS TO THE AUTHORS:

Editor's Comments:

Editor

Comments to the Author:

The reviewers made several important comments and suggestions. The authors have to address each reviewer comments in detail. Also authors have to properly justify the proposed method with respect to existing literature on this topic

Reviewers' Comments:

Reviewer: 1

Comments to the Author

The authors have done lots of work on transient stability analysis with TEF and tried to applied it to real time prediction and control. However, the reviewer has following questions:

1. The last paragraph of section I should be revised to reflect the content of the following sections.
2. The key point of this paper is the match of KEs from PMUs with offline calculation. But only stored and measured KEs are used, another indicator (such as fault location or rotor speed) should be included to guarantee the selected MOD is the right one.
3. An overall time cost evaluation should be included to show that the method can be used REAL TIME.
4. In table2, there are too many contingencies with too many generators in MODs. Generators, which tend to be out of step, are referred to as th MOD. It is unbelievable that all generators would be out of step. The results should be double-checked.
5. Another case study should be used to show the possible application of this method in large scale power system.
6. Regarding the advantages in section V, almost all of them are with the TEF itself instead of the novelty of this paper.

Reviewer: 2

Comments to the Author

IEEE Transactions on Power Systems

Review of the Paper TPWRS_00958_2015 "Real-Time Prediction and Control of Transient Stability Using Transient Energy Function"

This paper presents an interesting approach for computing transient stability margin based on look-up table of Modes of Disturbances (MODs). The main advantage of the proposed concept according to the authors is that it overcomes difficulties associated with real-time calculations of Controlling Unstable Equilibrium Points (CUEPs).

1. Generally speaking, any advancement in the area of the real-time, TEF based, transient stability control, especially if it could use existing PMU measurements, is of great interest. Consequently, I encourage the authors to continue developing their ideas in this area.
2. My concerns with this version of the paper would be primarily related to aspects of practical implementation, having in mind that the paper is oriented towards real-time control of transient stability. In large power systems (for sake of example let's consider the U.S Eastern Interconnection), with thousands of generators, would accuracy and applicability of the proposed method be challenged by a) lack of PMUs at majority of generator's locations b) computational complexity of calculating MODs and CUEPs for faults at each location of the grid c) estimation of post-fault topology based on limited number of PMUs at large systems d) varying unit commits/generation dispatch and d) variations of system loading conditions not only along daily loading curve but also due to seasonal loading variations.
3. The concept of "maximum sensitivity of the energy margin to generation shedding in different plants" (Page 4, Section III-c) should be better explained. Is this calculation being performed in a centralized manner or distributed among several regional control centers? How fast this sensitivity calculation needs to be (with re-calculation of $(V(t)-V_{cr})$), in order to allow time for feedback control signal to be sent to trip selected generator(s)? Did simulations detect any cases in which the proposed approach underestimated the generator shedding requirement (please expand your thoughts related to Page 6, Step 8, second column, paragraph 1)?

Reviewer: 3

Comments to the Author

The paper proposes a TEF based method to predict transient stability and mitigate the situation if the system is found to be vulnerable. At the beginning of the paper it is not clear whether the prediction is made before the event happens or after the event happens. This should be clearly stated at the beginning of the paper. In the second column of the first page under the point number (3), I believe that the authors mean "quantitative" not "qualitative" please check this.

If the power system is in a secure state, it does not go unstable for a single contingency. The scenario addressed in the paper is that a single contingency occur and the proposed method is used to determine the transient stability. This means that the power system was not in a secure state when the contingency occurred. The method proposed is for corrective control not preventive control. Please clarify these points.

The proposed method involves a heavy computational burden given the condition that the fault has already happened and a quick decision needs to be taken to make corrective actions. Authors have not addressed this critical issue.

The KE of the system after 0.1s is calculated in the proposed method. The rational of this choice is not clear. Does it depend on the system? There are standard PMU reporting rates. How do the authors suggest to obtain the data at exactly 0.1s instant. What is the effect of data transmission delays on the proposed method?

Authors claim that the other available methods have the disadvantage that the trained algorithms are dependent on the system topology, but the proposed method does not. However, the MOD is determined by using a nearest neighbour search technique. The data for this search are generated off-line and saved for real time use. Therefore the authors' claim is not valid.

The error of angle estimation is justified by citing a single reference where an error of 10 degrees is used as the acceptable value. This value is not a standard or well accepted value. The authors need to justify this choice for the particular application.

Please discuss how the proposed method is superior to the methods in references [19,20,24,25,26,27,29,and 30]. There are also methods that use the samples of voltage magnitude after fault clearance to predict transient stability. What is the advantage of the proposed method over that approach.

Reviewer: 4

Comments to the Author

The paper proposes a method to implement real time stability control using Transient Energy Function method, using PMU data. It is proposed to use a look-up table created using off-line simulations to overcome some of the problems associated with Energy Function method, such as the need for knowing the fault and post fault system configuration and out step generator groups. The need for simulation of post fault system is avoided by predicting the future rotor angles using a curve fitting approach.

Although the method is having some theoretical footing, in this reviewer's opinion, the method is quite complicated for real time implementation and prone to errors at many points, as somewhat evident from the results. How would this be much different from machine learning type approaches? The creation of off-line table requires consideration of different operating points and different faults, and selection of MOD still requires some classification type approach. Some of the machine learning type approaches has also shown that they can tolerate topology changes up to certain extent.

Step-3 in Section III-B requires calculation of post fault stable equilibrium point for several probable cases, and it is proposed to use the procedure in Step-1 in Section II-B, which is said to be a load flow like minimization problem. How would you apply this for a large system in real time?

Generator speed is not usually provided as a standard quantity in synchrophasor data. You need to physically measure the generator speed and include that as an analog data. Or else, the generator bus frequency (which is provided by PMUs) is used as an approximation. How critical is the generator speed for the accuracy of the algorithm?

Even the generator rotor angles are not directly measured, they need to be estimated from the terminal measurements and machine impedance parameters. Can you use the generator terminal voltage angles instead of rotor angles?

The presentation quality of the paper has suffered from the effort to keep it short. Lack of background make this paper difficult to understand for a person who is not very familiar with energy function methods. The theory presented in Section II-A is neither explained nor justified. Also no references to related work are provided. This section is very abrupt. The same comments as above are applicable to Section II-B.

Define Symbols before using them or immediately after (for ex. second last para in the introduction, what are ΔV_{pe} and KE_{corr} ?). Acronyms must be defined at the first use (for ex. CUEP in the abstract, BCU in introduction, etc.). As many abbreviations are used, it may be good to have a list of symbols and abbreviations at the beginning.

In equations (1) – (4) use of indices i and j are confusing. First it is mentioned that machines i and j are the unstable machines, but then i is used as an index in (2) and (4).

What is the minimizing function $f_i^{PF}(\theta)$?

Many of the variables in (5) and (6) are not defined. What are C_{ij} and D_{ij} ? What is E ?

The transient energy function in (9) is said to be well established, but no reference is given.

In the conclusion, it is claimed that the main contribution of the paper is "calculation of CUEP in real time". As I understand the MOD method of calculating CUEP presented in Section II-A is not new.

When you shed generators to avoid instability, how would you keep the load-generation balance in a practical system? You probably need some load shedding subsequent to the generator shedding. (In 39 bus test system, this is not visible as the lost generation usually supplied by the large equivalent generator)

In the introduction - "The unstable equilibrium points of the generators corresponding to the MOD and the stable equilibrium points of other generators form the Controlling Unstable Equilibrium Point (CUEP)." ..???? Is something missing here?

Reviewer: 5

Comments to the Author

This paper presents a real time control method for transient stability using PMU. I understand that the proposed method just follows the basic theory of the transient energy function method but it is difficult to find special new findings in the paper. The followings are to be considered by the authors.

(1) This reviewer think that the method is based on the well-known transient energy function method but the authors have not properly reviewed the past studies. This reviewer think that [A1] below should be cited. The method works in real power system for 20 years.

[A1] H. Ota, et al., "Development of transient stability control system (TSC system) based on on-line stability calculation," IEEE Trans on Power System, Vol. PWR-11, No. 3, pp. 1463-1472, 1996.

(2) The authors use index (dV_{pe}/KE_{corr}) to identify MOD in step 3 in section III. B a). The index appears several times in the paper before this part but is not clearly identified; the explanation is finally given after step 5. This reviewer think that this part is not original but important for this paper, and should be more explained using equations. In general, the numerator (dV_{pe}) and denominator (KE_{corr}) show different performance, and it is questionable if the index shows an expected performance. I think that KE_{corr} may work better than dV_{pe} , but both of them may contain unacceptable errors depending on power system conditions. The authors should explain how the identification of MOD is correctly carried out from the point of view of the individual factors using more analysis data.

(3) The explanation of (KE_{corr}) after step 5 above shows that the paper use the concept of the two machine equivalent. If so, the authors should refer to the research study by M. Pavella, et al. I think that the proposed method is similar to this work as a result.

(4) I think that the most critical part of the proposed method is "Estimation of stability" in section III. B. b) and c). This part is theoretically acceptable but the computation models themselves are erroneous and require more reliability to be used in real system. The first factor is that the stability judgement by the energy function itself is erroneous, even when the accurate CUEP is obtained using reliable method. Therefore, its use is usually limited to screening purpose before the final stability judgement.

(5) The control method based on the energy function may be effective in many cases in my experience but cannot be useful unless the accuracy is guaranteed. The authors should add information about the accuracy of the proposed method compared with the exact numerical simulations. For example, the readers must be interested in the accuracy of the total amount of generator or load shedding, which should be compared between the proposed method and the conventional numerical simulations.

(6) PMU data is also known to be unreliable, and therefore, it may worsen the accuracy of the proposed control method. Although I basically interested in the transient stability

control method, the authors should more focus on the accuracy issues.
(7) All reference papers should be cited in the main text.

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